

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S8 (S) Examination September 2025 (2019 Scheme)

Course Code: CXT434**Course Name: SOFTWARE TESTING AND QUALITY ASSURANCE****Max. Marks: 100****Duration: 3 Hours****PART A***Answer all questions, each carries 3 marks.*

Marks

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|----|---|-----|
| 1 | What are the main objectives of software testing? | (3) |
| 2 | Define verification and validation in testing phase. | (3) |
| 3 | What is the role of a "stub" and a "driver" in integration testing? | (3) |
| 4 | Differentiate between aesthetic testing and accessibility testing. | (3) |
| 5 | What is quality standards? | (3) |
| 6 | Explain the concept of six sigma and its significance in process improvement. | (3) |
| 7 | Explain the process of dynamic slicing. | (3) |
| 8 | What is a test plan? | (3) |
| 9 | Explain the key benefits and challenges of test automation. | (3) |
| 10 | Explain the principles of object-oriented testing. | (3) |

PART B*Answer any one full question from each module, each carries 14 marks.***Module I**

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|----|---|-----|
| 11 | a) Define software testing. What is the role of testing in ensuring software quality? | (7) |
| | b) Differentiate between static analysis and dynamic analysis. | (7) |

OR

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| 12 | a) Explain the five views of software quality. How do these views contribute to understanding and managing software quality? | (14) |
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Module II

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|----|---|-----|
| 13 | a) Explain the different types of white box testing techniques and how can they be classified based on their focus and application in software testing? | (8) |
| | b) Define usability testing. Explain its importance in software development and user experience design. | (6) |

OR

- 14 a) Explain the importance of testing at boundary values in software systems. How do these tests help in uncovering defects that might not be found by random or general testing? (14)

Consider an online voting system where the age must be between 18 and 120 years. Using boundary value analysis, create test cases to validate this input field.

Module III

- 15 a) Sketch a diagram of the five maturity levels in the capability maturity model. Discuss the progression of an organization through these levels, highlighting the key characteristics and improvements associated with each level. (14)

OR

- 16 a) Explain the eight principles of the ISO 9000:2000 software quality standard. (8)
b) Discuss the relationship between quality factor and criteria. (6)

Module IV

- 17 a) Explain the role of selenium in regression testing. Discuss its features, advantages and how it can be used to automate regression test cases. (7)
b) Write a short note on tools for regression testing. (7)

OR

- 18 a) Discuss the role of test reporting in regression testing and explain how different types of test reports help stakeholders assess testing progress and software quality. (14)

Module V

- 19 a) Draw a diagram representing the key components of test automation. Explain the role of each component in the automation framework and how they work together to ensure efficient test execution. (8)
b) Explain the different types of software quality metrics. (6)

OR

- 20 a) Describe the characteristics of conventional algorithm-centric programming languages and explain how they focus on algorithm development in software programming. (14)
